

Principles of Microeconomics — Day 2

In-Class Problem Set: Perfect Competition

This problem set accompanies the Day 2 session. Numbers differ from the lecture example. Solutions (instructor copy) are collected at the end; set `\solnfalse` in the preamble to produce a student version.

Exercise 1 — Market equilibrium and the long run

A competitive market consists of identical firms, each with short-run total cost

$$TC(q) = q^2 + 2q + 25,$$

and market demand $Q^D = 1300 - 50P$. In the short run there are 100 firms.

- (a) Find each firm's MC and AVC , its shutdown price, and its supply $q(P)$.
- (b) Write short-run market supply, then find the short-run equilibrium price and quantity, each firm's output, and each firm's profit.
- (c) Is profit positive? What does this set in motion?
- (d) In the long run, find $\min ATC$, the long-run price, each firm's output, and the number of firms.

Exercise 2 — One firm's operate-or-shut-down decision

A single competitive firm has short-run total cost

$$TC(q) = q^3 - 8q^2 + 30q + 50.$$

- (a) Find MC and AVC . Show that AVC is minimised at $q = 4$ and use this to state the shutdown price.
- (b) The market price is $P = 42$. Find the firm's profit-maximising output and its profit. What does a positive profit signal for the market?
- (c) The price falls to $P = 14$. Find the firm's output and its profit at this price, and compare that profit with what it would earn by shutting down. What is special about this price?
- (d) The price falls further to $P = 10$. What does the firm do, and what is its profit (or loss)?